

Wren Oswin

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SUMMARY

Software Engineer and MTech candidate at IIT Kanpur with industry experience in OTT application development and a background in computer vision research. Skilled in JavaScript, Python, and C/C++, with a track record of delivering production features, improving test coverage, and building cross-platform systems.

EDUCATION

Indian Institute of Technology, Kanpur <i>Master of Technology in Computer Science and Engineering</i>	Kanpur, Uttar Pradesh July 2026 – Present
Sree Chitra Thirunal College of Engineering <i>Bachelor of Technology in Computer Science and Engineering CGPA: 9.31/10</i>	Thiruvananthapuram, Kerala Aug. 2019 – July 2023
Believer's Church Residential School <i>Senior Secondary Grade: 93.6%</i>	Thiruvalla, Kerala June 2017 – March 2019

EXPERIENCE

Software Engineer <i>Tata Elxsi Ltd.</i>	July 2023 – January 2025 Thiruvananthapuram, Kerala
- Delivered 8–10 UI feature tickets on a production OTT application using JavaScript, LightningJS, and MongoDB, improving playback performance and UI responsiveness for end users - Wrote unit tests for 5–10 modules across the codebase, driving overall test coverage from 50% to 80% and significantly reducing untested surface area ahead of releases - Resolved 10–15 testing and bug-fix tickets within Agile sprints, including log analysis and graphical testing to diagnose performance issues in live applications - Contributed to a confidential POC using React Native and Shaka Player — rewrote core playback logic and resolved cross-platform dependency conflicts to advance the port of an OTT application to a new OS	
Software Engineer Intern <i>Tata Elxsi Ltd.</i>	Sep. 2022 – June 2023 Thiruvananthapuram, Kerala
- Gained hands-on experience with DevOps tooling — including Terraform, Ansible, Docker, and Concourse — to build and manage CI/CD pipelines - Applied Agile methodologies in a team environment across the full SDLC, from requirements gathering to deployment - Built foundational knowledge of AWS Cloud services, including compute (EC2), storage (S3), and deployment workflows	

PROJECTS

Visualizing Floor Plans Using Virtual Reality <i>Python, OpenCV, Blender, PyQt5</i>	June 2022 – June 2023
- Built an image preprocessing pipeline using OpenCV (grayscale conversion, adaptive thresholding, morphological filtering) to extract structural elements from 2D architectural floor plans with high accuracy - Implemented automated door and window detection using ORB keypoint extraction and Brute-Force feature matching, enabling reliable tagging of architectural components without manual annotation - Engineered a contour detection and segmentation system to convert processed floor plan images into 3D Blender models, forming the basis for an interactive VR visualization tool	

TECHNICAL SKILLS

Languages: JavaScript, Python, C/C++, Java, Dart, SQL (PostgreSQL), HTML/CSS, R
Frameworks & Libraries: React Native, LightningJS, Flutter, Node.js, Express.js, Jest, Cypress, pandas, NumPy, Matplotlib
Developer Tools: Git, Docker, Terraform, Ansible, AWS (EC2, S3), VS Code, Jupyter
Operating Systems: Linux, Windows, macOS